Hands-on Activity 0.2	
Hands-on Activity 0.2 - I/O read and write	
Course Code: CPE 009B	**Program:** Computer Engineering
Course Title: Object Oriented Programming	**Date Performed: 07-18-26**
Section: CPE21S2	**-Date Submitted: 07-18-26**
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A. Instructions

Tasks

1. **Perform the laboratory activity using Google Colab. Place your code in the specified programs. Laboratory activity: CPE 009 Lab Activity 2 - Inheritance, Encapsulation, and Abstraction.pdfDownload CPE 009 Lab Activity 2 - Inheritance, Encapsulation, and Abstraction.pdf**
2. **Document your lab activity properly and observe correct English grammar and formatting.**
3. **Answer all the supplementary activities (programs and questions).**
4. **Write your conclusion**
5. **Submit here the .pdf file with the file name: cpe009b_lastname_fistname_lab2.pdf**

Note: Submission beyond the due date with no valid excuse letter / valid reason will not be accepted.

B. Output(s) and Observation(s)

```
class Character (): #Starts the blueprint for a character.
    def __init__(self, username): # Runs when you make a new character.
        self.__username = username #  Sets a private name.
        self.__hp = 100 # Starts life points at 100.
        self.__mana = 100 # Starts magic points at 100.
        self.__damage = 5 # Sets base attack damage to 5.
        self.__str = 0 # Starts strength at zero.
        self.__vit = 0 # Starts vitality at zero
        self.__int = 0 # Starts intelligence at zero.
        self.__agi = 0 #  Starts agility at zero.
    def getUsername(self):
        return self.__username # Returns the private name.
    def setUsername(self, new_username):
        self.__username = new_username #Returns the private name.
    def getHp(self):
        return self.__hp #Returns current life points.
    def setHp(self, new_hp):
        self.__hp = new_hp #Changes life points.
    def getDamage(self):
        return self.__damage #Returns attack damage.
```

```
    def setDamage(self, new_damage):
        self.__damage = new_damage # Changes attack damage.
    def getStr(self):
        return self.__str # Returns strength value.
    def setStr(self, new_str):
        self.__str = new_str #Changes strength value.
    def getVit(self):
        return self.__vit #Returns vitality value.
    def setVit(self, new_vit):
        self.__vit = new_vit #Changes vitality value.
    def getInt(self):
        return self.__int #Returns intelligence value.
    def setInt(self, new_int):
        self.__int = new_int #Changes intelligence value.
    def getAgi(self):
        return self.__agi#Returns agility value.
    def setAgi(self, new_agi):
        self.__agi = new_agi # changes agility value
    def reduceHp(self, damage_amount):
        self.__hp = self.__hp - damage_amount# subracts damage from life points
    def addHp(self, heal_amount):
        self.__hp = self.__hp + heal_amount#adds healing to life points

character1 = Character("Your Username")#creates a new character object
print (f"character1._username")# prints literal text
print (character1.getUsername())#calles the getter method and prints the correct
username
```

```
class Novice(Character): # creates a subclass that inherits all methods and
attributes from character
    def basicAttack (self, character): #defines a method to attack another character
target
        character.reduceHp(self.getDamage()) #lowers the target's hp using the
attacker's damage value
        print (f"{self.getUsername()} performed Basic Attack! -{self.getDamage()}")
#prints the combat action text to the screen

character1 = Novice("Your Username") #creates a new novice player
print (character1.getUsername()) #calls the inherited getter and print the username
print (character1.getHp()) # calls the inherited getter and prints the default
starting hp
```

```
class Swordsman(Novice): #creates a subclass inheriting from novice
    def __init__(self,username): # Defines the constructor method
        super().__init__(username)#  A commented-out line that would normally call the parent class constructor.
        self.setStr(5) #Sets the swordsman's strength value to 5 using an inherited method.
        self.setVit(10)#Sets the swordsman's vitality value to 10 using an inherited method.
        self.setHp(self.getHp()+self.getVit()) #Increases the character's total HP by adding their current HP and vitality value.

    def slashAttack(self, character): #Defines an action method that takes another character as a target.
        self.new_damage = self.getDamage()+self.getStr() #Calculates attack damage by adding base damage and strength.
        character.reduceHp(self.new_damage) #Calls a method on the target character to subtract the calculated damage from their HP.
        print (f"{self.getUsername()} performed Slash Attack! -{self.new_damage}") #Prints an action message showing who attacked and how much damage was dealt.
```

```
import random #generate random numbers.

class Archer(Novice):
    def __init__(self, username): #Defines the constructor method to initialize an Archer object.
        super().__init__(username) # Calls the parent Novice constructor to properly register the username.
        self.setAgi(5) #Sets the character's agility stat to 5.
        self.setInt(5)#Sets the character's intelligence stat to 5.
        self.setVit(5) #Sets the character's vitality stat to 5.
        self.setHp(self.getHp()+self.getVit()) #Adds the vitality value to the character's baseline health pool.

    def rangedAttack(self,character): #Defines the attack method taking a target character.
        self.new_damage = self.getDamage()+random.randint(0,self.getInt()) #Calculates damage by adding base damage to a random bonus roll between 0 and intelligence.
        character.reduceHp(self.new_damage) #apply damage
        print(f"{self.getUsername()} performed Ranged Attack! -{self.new_damage}")
```

```
class Magician(Novice): #Starts a new class
```

```
    def __init__(self, username): #Defines the constructor method to initialize a
Magician object.
        super().__init__(username) #Calls the parent Novice constructor to correctly
bind the username.
        self.setInt(10) #Sets the magician's core intelligence stat to 10.
        self.setVit(5) #Sets the magician's baseline vitality stat to 5.
        self.setHp(self.getHp()+self.getVit()) #Boosts total starting health by
adding the vitality value to current HP.

    def heal(self): #Defines a self-recovery action method that takes no external
target.
        self.addHp(self.getInt()) #Restores the magician's health points by an
amount equal to their intelligence stat.
        print(f"{self.getUsername()} performed Heal! +{self.getInt()}") #Outputs a
combat log message tracking the character name and health restored.
    def magicAttack(self,character): #Defines an offensive spellcasting method
targeting an enemy character object.
        self.new_damage = self.getDamage()+self.getInt() #Calculates magic damage by
adding intelligence directly to base damage.
        character.reduceHp(self.new_damage) #Subtracts the calculated spell damage
from the target character's health.
        print(f"{self.getUsername()} performed Magic Attack! -{self.new_damage}")
#Outputs a combat log tracking who cast the spell and how much damage was dealt.
```

```
class Boss(Swordsman, Archer, Magician): #Defines a Boss class inheriting attributes
and skills from all three subclasses.
    def __init__(self, username): # Defines the constructor method to initialize the
ultimate boss entity.
        super().__init__(username)

        self.setStr(10) #Sets the boss's physical strength multiplier to 10.
        self.setVit(25) #Sets a high boss-level vitality floor of 25.
        self.setInt(5) # Sets the boss's magical intelligence factor to 5.
        self.setHp(self.getHp() + self.getVit()) #Boosts the boss's health pool
substantially using its massive vitality stat.
```

```
#test.py
Character1 = Swordsman("Royce")
Character2 = Magician("Archie")
print(f"{Character1.getUsername()} HP: {Character1.getHp()}")
print(f"{Character2.getUsername()} HP: {Character2.getHp()}")
Character1.slashAttack(Character2)
Character1.basicAttack(Character2)
```

```
print(f"{Character1.getUsername()} HP: {Character1.getHp()}")
print(f"{Character2.getUsername()} HP: {Character2.getHp()}")
Character2.heal()
Character2.magicAttack(Character1)
print(f"{Character1.getUsername()} HP: {Character1.getHp()}")
print(f"{Character2.getUsername()} HP: {Character2.getHp()}")
```

```
Royce HP: 110
Archie HP: 105
Royce performed Slash Attack! -10
Royce performed Basic Attack! -5
Royce HP: 110
Archie HP: 90
Archie performed Heal! +10
Archie performed Magic Attack! -15
Royce HP: 95
Archie HP: 100
```

```
import random

class Game:
    def __init__(self):
        self.p1_wins = 0
        self.p2_wins = 0

    def select_role(self, name, allow_advanced=True):
        if not allow_advanced:
            return Novice(name)

        print(f"\nSelect a role for {name}:")
        print("1. Swordsman\n2. Archer\n3. Mage\n4. Novice")
        choice = input("Enter choice (1-4): ")

        if choice == "1": return Swordsman(name)
        if choice == "2": return Archer(name)
        if choice == "3": return Magician(name)
        return Novice(name)

    def combat_loop(self, player1, player2):
        print(f"\n--- MATCH START: {player1.getUsername()} vs
{player2.getUsername()} ---")

        while player1.getHp() > 0 and player2.getHp() > 0:
            fighters = [player1, player2]
            random.shuffle(fighters)
```

```
            attacker = fighters[0]
            defender = fighters[1]

            print(f"\n{attacker.getUsername()}'s turn!")

            available_skills = []
            if hasattr(attacker, 'slashAttack'):
available_skills.append(attacker.slashAttack)
            if hasattr(attacker, 'magicAttack'):
available_skills.append(attacker.magicAttack)
            if hasattr(attacker, 'rangedAttack'):
available_skills.append(attacker.rangedAttack) # Fixed "d" typo

            if available_skills:
                chosen_skill = random.choice(available_skills)
                chosen_skill(defender)
            else:
                attacker.basicAttack(defender)

            print(f"{defender.getUsername()} HP: {defender.getHp()}")

            if defender.getHp() <= 0:
                print(f"\n{defender.getUsername()} has been defeated!")
                return attacker

    def start(self):
        while True:
            print("\n==== GAME MENU ====")
            print("1. Single Player (vs Boss)")
            print("2. Player vs Player")
            print("3. Exit")
            mode = input("Select Mode: ")

            if mode == "1":
                # Requirement: Start as Novice
                role_unlocked = self.p1_wins >= 2
                p1 = self.select_role("Player 1", allow_advanced=role_unlocked)
                boss = Boss("Monster")

                winner = self.combat_loop(p1, boss)
                if winner == p1:
                    self.p1_wins += 1
                    print(f"Total Wins: {self.p1_wins}")
```

```
            elif mode == "2":
                p1 = self.select_role("Player 1")
                p2 = self.select_role("Player 2")

                winner = self.combat_loop(p1, p2)
                if winner == p1: self.p1_wins += 1
                else: self.p2_wins += 1
                print(f"Scores -> P1: {self.p1_wins} | P2: {self.p2_wins}")

            elif mode == "3":
                break

if __name__ == "__main__":
    my_game = Game()
    my_game.start()
```

```
==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 1

--- MATCH START: Player 1 vs Monster ---

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 140

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 135

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 130

Monster's turn!
Monster performed Slash Attack! -15
Player 1 HP: 85

Monster's turn!
Monster performed Slash Attack! -15
Player 1 HP: 70

Monster's turn!
Monster performed Ranged Attack! -6
Player 1 HP: 64
```

```
Monster performed Magic Attack! -10
Player 1 HP: 54
...

Monster's turn!
Monster performed Magic Attack! -10
Player 1 HP: 44

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 125

Monster's turn!
Monster performed Ranged Attack! -8
Player 1 HP: 36

Monster's turn!
Monster performed Magic Attack! -10
Player 1 HP: 26

Monster's turn!
Monster performed Magic Attack! -10
Player 1 HP: 16

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 120

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 115

Monster's turn!
Monster performed Ranged Attack! -7
Player 1 HP: 9
```

```
Monster's turn!
Monster performed Ranged Attack! -7
Player 1 HP: 2

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 110

Player 1's turn!
Player 1 performed Basic Attack! -5
Monster HP: 105

Monster's turn!
Monster performed Magic Attack! -10
Player 1 HP: -8

Player 1 has been defeated!

==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 1

Select a role for Player 2:
1. Swordsman
2. Archer
```

```
    2. Archer
    3. Mage
    4. Novice
    Enter choice (1-4): 1

    --- MATCH START: Player 1 vs Player 2 ---

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 100

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 90

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 80

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 70

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 60

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 50

    Player 1's turn!
    Player 1 performed Slash Attack! -10
    Player 2 HP: 100
```

```
Player 2's turn!
Player 2 performed Slash Attack! -10
Player 1 HP: 40

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 90

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 80

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 70

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 60

Player 2's turn!
Player 2 performed Slash Attack! -10
Player 1 HP: 30

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 50

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 40

Player 2's turn!
Player 2 performed Slash Attack! -10
```

```
Player 2 performed Slash Attack! -10
Player 1 HP: 20

...
Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 30

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 20

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 10

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 0

Player 2 has been defeated!
Scores -> P1: 1 | P2: 0

==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 1
```

```
Select a role for Player 2:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 2

--- MATCH START: Player 1 vs Player 2 ---

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 95

Player 2's turn!
Player 2 performed Ranged Attack! -8
Player 1 HP: 102

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 85

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 75

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 65

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 55

Player 2's turn!
```

```
Player 2's turn!
Player 2 performed Ranged Attack! -10
Player 1 HP: 92

Player 2's turn!
Player 2 performed Ranged Attack! -6
Player 1 HP: 86

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 45

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 35

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 25

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 15

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 5

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: -5

Player 2 has been defeated!
Scores -> P1: 2 | P2: 0
```

```
==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 1

Select a role for Player 2:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 3

--- MATCH START: Player 1 vs Player 2 ---

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 95

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 95

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 85

Player 1's turn!
```

```
Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 75

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 65

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 55

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 80

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 45

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 65

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 50

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 35

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 25
```

```
Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 15

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 5

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 20

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 5

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: -10

Player 1 has been defeated!
Scores -> P1: 2 | P2: 1

==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
```

```
    4. Novice
    Enter choice (1-4): 1
...
    Select a role for Player 2:
    1. Swordsman
    2. Archer
    3. Mage
    4. Novice
    Enter choice (1-4): 4

    --- MATCH START: Player 1 vs Player 2 ---

    Player 1's turn!
    Player 1 performed Slash Attack! -10
    Player 2 HP: 90

    Player 2's turn!
    Player 2 performed Basic Attack! -5
    Player 1 HP: 105

    Player 1's turn!
    Player 1 performed Slash Attack! -10
    Player 2 HP: 80

    Player 1's turn!
    Player 1 performed Slash Attack! -10
    Player 2 HP: 70

    Player 1's turn!
    Player 1 performed Slash Attack! -10
    Player 2 HP: 60

    Player 2's turn!
    Player 2 performed Basic Attack! -5
    Player 1 HP: 100
```

```
Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 85

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 80

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 75

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 70

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 30

Player 1's turn!
Player 1 performed Slash Attack! -10
Player 2 HP: 20

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 65

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 60

Player 1's turn!
Player 1 performed Slash Attack! -10
```

```
    Player 1 performed Slash Attack! -10
    Player 2 HP: 10
...
    Player 1's turn!
    Player 1 performed Slash Attack! -10
    Player 2 HP: 0

    Player 2 has been defeated!
    Scores -> P1: 3 | P2: 1

    ==== GAME MENU ====
    1. Single Player (vs Boss)
    2. Player vs Player
    3. Exit
    Select Mode: 2

    Select a role for Player 1:
    1. Swordsman
    2. Archer
    3. Mage
    4. Novice
    Enter choice (1-4): 2

    Select a role for Player 2:
    1. Swordsman
    2. Archer
    3. Mage
    4. Novice
    Enter choice (1-4): 3

    --- MATCH START: Player 1 vs Player 2 ---

    Player 2's turn!
    Player 2 performed Magic Attack! -15
    Player 1 HP: 90
```

```
Player 1 HP: 90

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 75

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 60

Player 1's turn!
Player 1 performed Ranged Attack! -7
Player 2 HP: 98

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 45

Player 1's turn!
Player 1 performed Ranged Attack! -5
Player 2 HP: 93

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 30

Player 1's turn!
Player 1 performed Ranged Attack! -6
Player 2 HP: 87

Player 1's turn!
Player 1 performed Ranged Attack! -10
Player 2 HP: 77

Player 2's turn!
```

```
Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 15
...

Player 1's turn!
Player 1 performed Ranged Attack! -5
Player 2 HP: 72

Player 1's turn!
Player 1 performed Ranged Attack! -6
Player 2 HP: 66

Player 2's turn!
Player 2 performed Magic Attack! -15
Player 1 HP: 0

Player 1 has been defeated!
Scores -> P1: 3 | P2: 2

==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 2

Select a role for Player 2:
1. Swordsman
2. Archer
```

```
3. Mage
4. Novice
Enter choice (1-4): 4

--- MATCH START: Player 1 vs Player 2 ---

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 100

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 95

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 90

Player 1's turn!
Player 1 performed Ranged Attack! -6
Player 2 HP: 94

Player 1's turn!
Player 1 performed Ranged Attack! -5
Player 2 HP: 89

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 85

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 80
```

```
Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 75

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 70

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 65

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 60

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 55

Player 1's turn!
Player 1 performed Ranged Attack! -9
Player 2 HP: 80

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 50

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 45

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 40
```

```
Player 1 HP: 40

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 35

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 30

Player 1's turn!
Player 1 performed Ranged Attack! -7
Player 2 HP: 73

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 25

Player 1's turn!
Player 1 performed Ranged Attack! -6
Player 2 HP: 67

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 20

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 15

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 10

Player 2's turn!
```

```
Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 5

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 0

Player 1 has been defeated!
Scores -> P1: 3 | P2: 3

==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 3

Select a role for Player 2:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 4

--- MATCH START: Player 1 vs Player 2 ---

Player 2's turn!
Player 2 performed Basic Attack! -5
```

```
Player 2 performed Basic Attack! -5
Player 1 HP: 100
...
Player 1's turn!
Player 1 performed Magic Attack! -15
Player 2 HP: 85

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 95

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 90

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 85

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 80

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 75

Player 1's turn!
Player 1 performed Magic Attack! -15
Player 2 HP: 70

Player 1's turn!
Player 1 performed Magic Attack! -15
Player 2 HP: 55
```

```
Player 1's turn!
Player 1 performed Magic Attack! -15
Player 2 HP: 40

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 70

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 65

Player 1's turn!
Player 1 performed Magic Attack! -15
Player 2 HP: 25

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 60

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 55

Player 2's turn!
Player 2 performed Basic Attack! -5
Player 1 HP: 50

Player 1's turn!
Player 1 performed Magic Attack! -15
Player 2 HP: 10

Player 1's turn!
Player 1 performed Magic Attack! -15
```

```
Player 2 HP: -5

Player 2 has been defeated!
Scores -> P1: 4 | P2: 3

==== GAME MENU ====
1. Single Player (vs Boss)
2. Player vs Player
3. Exit
Select Mode: 2

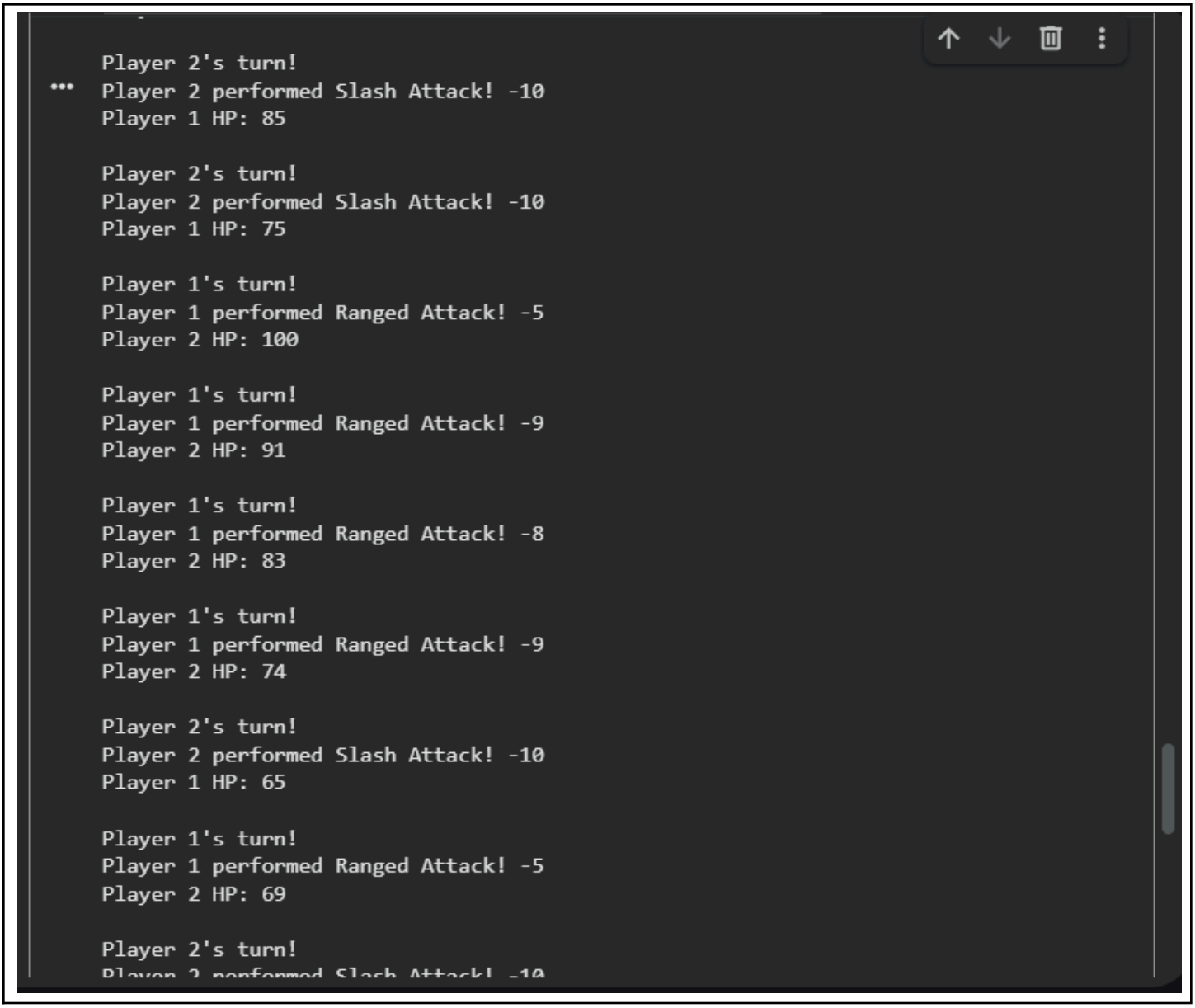
Select a role for Player 1:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 2

Select a role for Player 2:
1. Swordsman
2. Archer
3. Mage
4. Novice
Enter choice (1-4): 1

--- MATCH START: Player 1 vs Player 2 ---

Player 1's turn!
Player 1 performed Ranged Attack! -5
Player 2 HP: 105

Player 2's turn!
Player 2 performed Slash Attack! -10
Player 1 HP: 95
```


Player 2's turn!
Player 2 performed Slash Attack! -10
Player 1 HP: 85
Player 2's turn!
Player 2 performed Slash Attack! -10
Player 1 HP: 75
Player 1's turn!
Player 1 performed Ranged Attack! -5
Player 2 HP: 100
Player 1's turn!
Player 1 performed Ranged Attack! -9
Player 2 HP: 91
Player 1's turn!
Player 1 performed Ranged Attack! -8
Player 2 HP: 83
Player 1's turn!
Player 1 performed Ranged Attack! -9
Player 2 HP: 74
Player 2's turn!
Player 2 performed Slash Attack! -10
Player 1 HP: 65
Player 1's turn!
Player 1 performed Ranged Attack! -5
Player 2 HP: 69
Player 2's turn!

```
    Player 2 HP: 69

•••  Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 55

    Player 1's turn!
    Player 1 performed Ranged Attack! -7
    Player 2 HP: 62

    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 45

    Player 1's turn!
    Player 1 performed Ranged Attack! -9
    Player 2 HP: 53

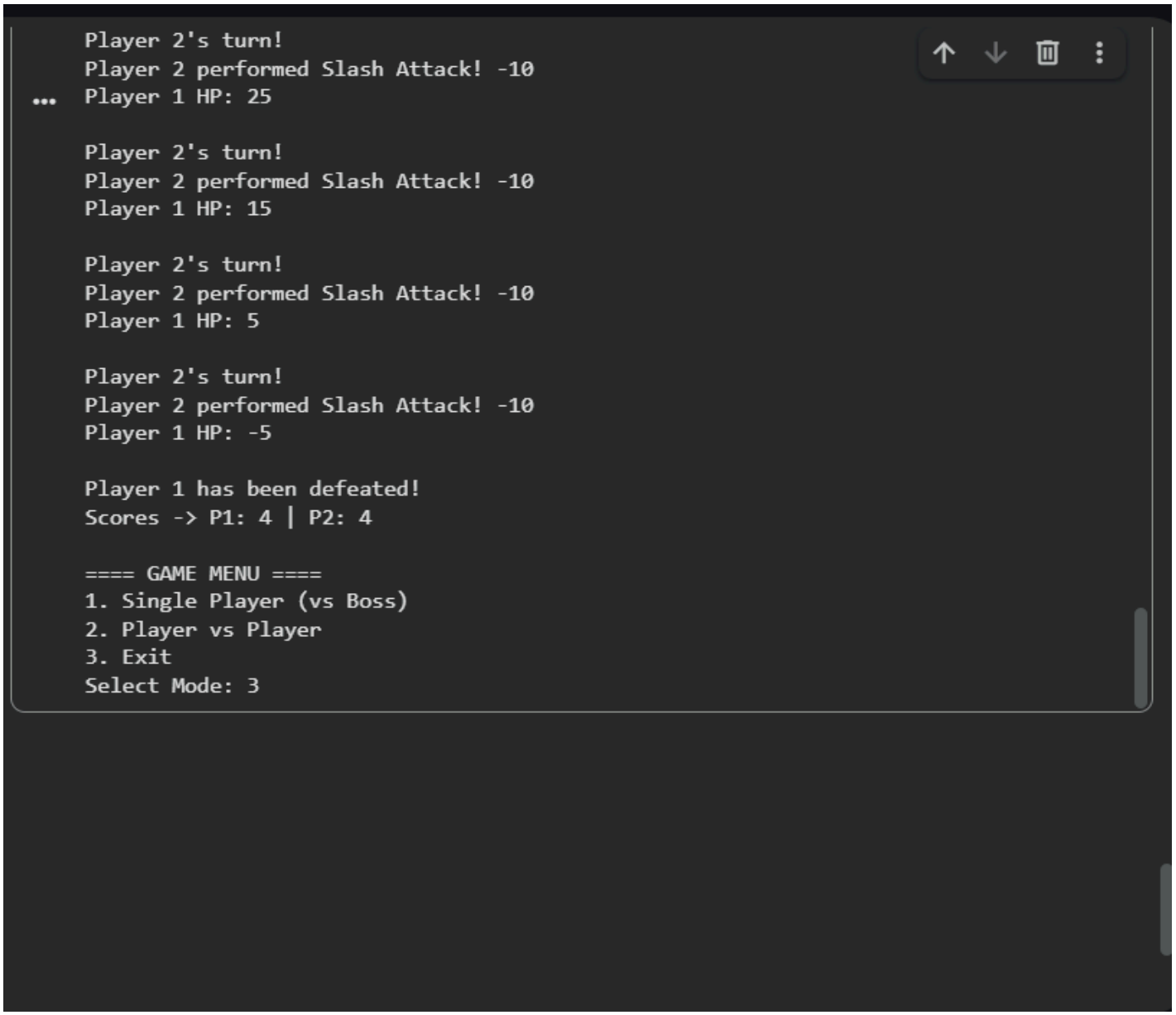
    Player 2's turn!
    Player 2 performed Slash Attack! -10
    Player 1 HP: 35

    Player 1's turn!
    Player 1 performed Ranged Attack! -5
    Player 2 HP: 48

    Player 1's turn!
    Player 1 performed Ranged Attack! -7
    Player 2 HP: 41

    Player 1's turn!
    Player 1 performed Ranged Attack! -10
    Player 2 HP: 31

    Player 2's turn!
```



Google Colab Link:
https://colab.research.google.com/drive/1_YPnqh7ZdQeF4uqqRDijAuvIs_MztnKN#scrollTo=QPAG_xeohZJ4

C. Conclusion & Lessons Learned

In this activity I have learned how to use classes, even if they are different classes are called, that is where encapsulation works. Inheritance work when I created a hierarchy, where there is a parent class and a child classes. Polymorphism works where different objects are interchangeable, by calling unique attacks, the game demonstrates how the classes will respond. In my conclusion, inheritance saves time to avoid duplication of codes, in short it can be reused per classes. Clean codes prevent crashes, typoes like miscapitalized typos or missing character can easily destroys the whole code, so at this point I have to be very carefully and look carefully. And lastly, combining multiple classes into a single class highlights python's internal interface also for it's complex software.